Plant Disease Classification (Vision)

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ABSTRACT

Plant disease is a common problem for home plant owners. This project proposes a lightweight plant disease classification system based on images. The system is based on a fine-tuned Convolutional Neural Network (CNN) architecture named MobileNet. The system is developed to help home plant owners detect various plant leaf diseases. The system will be based on images taken with a smartphone. To ensure the accuracy of the results, the model is trained on a large number of healthy and diseased plant leaf images. Data augmentation is also performed to ensure the accuracy of the results. The system will also give users feedback on the quality of the images. The system will also be available for farmers to use, as their experience may be greater than the AI model. The system is developed to give accurate results to users in an efficient manner to avoid further damage to the plants.

1. INTRODUCTION

Home care of plants is becoming a popular hobby, but one of the challenges that hobbyists face is plant diseases. Early detection of the signs of plant diseases is critical to ensure that the plants are healthy. Traditionally, plant disease detection is done by observation and expertise, but not all hobbyists have access to such expertise.

Recent advances in Artificial Intelligence, particularly in deep learning, provide a solution for this problem. Convolutional Neural Networks (CNNs) have demonstrated good performance in image-based classification problems, including plant disease detection problems. This project is based on designing a simple system that allows home plant owners to identify plant diseases based on plant leaf images. The system is based on a lightweight version of MobileNet, trained on a large dataset of healthy and diseased plant leaves.

The system helps users capture clear images, evaluates the image quality, and provides the predicted disease. Even though the tool is intended for home users, it has the option for usage by farmers in case they need immediate results on a larger scale. This project demonstrates the integration of AI technology and user interfaces for the development of user-friendly plant care solutions that help prevent plants from dying.

2. PROBLEM STATEMENT & SCOPE

2.1 PROBLEM STATEMENT

Home plant owners often struggle to identify diseases in their plants, leading to poor plant care or loss. While farmers usually have experience to detect plant problems, they may still benefit from a quick, automated check for large-scale crops. There is a need for an easy-to-use system that can classify plant leaf diseases from images, helping home users take better care of their plants and providing optional support to farmers.

2.2 SCOPE

The system will:

- Accept leaf images from home plant owners and classify them as healthy or diseased.
- Provide optional advisory support for farmers who may want a quick check on large-scale crops.
- Focus on common home and small-scale crops, using a lightweight MobileNet CNN model.
- Run on a smartphone or computer without heavy computation.

The system will not:

- Replace farmers' expert judgment for large-scale crop management.
- Diagnose diseases on stems, flowers, or fruits.
- Offer chemical prescriptions or complete treatment plans.

2.3 SUCCESS

- **Accuracy – Correct classification of leaf images.**
- **Macro F1 Score** – Balanced performance across disease types.
- **Confusion Matrix** – Identification of commonly confused disease classes.
- **User Feedback** – Home users find it easy and helpful; optional feedback from farmers.

2.4 METRICS

- Accuracy depends on the quality of images taken by users.
- Limited dataset may not cover all plant types.
- Processing speed may vary on lower-end devices.
- Ethical handling of user-submitted images must be ensured.

3. REVIEW OF RELATED LITERATURE

[1] In recent times, the application of deep learning-based models has yielded impressive results in the field of plant disease classification. Garg and Singh (2023) proposed the DeepPlantNet architecture as an efficient CNN-based framework that aims to classify various types of plant leaf diseases using improved computing performance and accuracy. The proposed model showed impressive performance in various classification scenarios, including eight-class, six-class, and three-class disease recognition. The proposed method was also able to learn discriminative features while using fewer parameters compared to the conventional CNN model. This demonstrates the recent trend of optimizing the performance of deep learning-based networks in the field of plant disease diagnosis.

[2] Further research has been carried out to compare the outcomes of the CNN models in classifying plant diseases. Saleem et al., (2020) performed an in-depth comparative study on various deep learning models like AlexNet, VGG-16, ResNet-50, DenseNet, MobileNet, and Xception using the same PlantVillage dataset. The purpose of this study is to assess the effect of various optimizers on deep learning models. From the results, it was clear that the results from the Xception model combined with the Adam optimizer were promising, with an accuracy of 99.81% and an F1 score of 0.9978. From these results, it is clear that CNN models can improve plant disease classification accuracy.

[3] Pal, Karmakar, & Mukherjee (2025) have developed a lightweight and explainable deep CNN-based model, referred to as Mob-Res, for plant disease image classification. This model utilized residual blocks and MobileNetV2 as a feature extractor. It was trained and validated on large-scale benchmark datasets, including Plant Disease Expert and PlantVillage, and produced high accuracy up to 99.47%. At the same time, it had fewer parameters, making it suitable for deployment on mobile devices. This research also utilized Explainable AI methods like Grad-CAM, Grad-CAM++, and LIME, enabling image regions that affect model predictions to be visualized.

[4] Mishra, Pandey, & G L (2025) conducted a comparative analysis on various deep learning architectures used for plant diseases using leaf images from potato and mango plant leaves. In this research, various convolutional neural networks were implemented and tested on various image datasets related to plant diseases. These networks include CNN, AlexNet, ResNet, and EfficientNet. The research used various image processing techniques to improve the overall accuracy of the model. Based on the research carried out, modern deep learning networks were able to outperform CNN networks in plant diseases using images. This is due to the feature extraction capabilities of modern networks.

4. DATASET PLAN

4.1 DATASET SOURCE

The project will primarily rely on the PlantVillage dataset for training and evaluating the plant disease classification model. The dataset contains more than 50,000 labeled images of healthy and diseased plant leaves, including various types of crops and plant diseases. The images were obtained under controlled environments and are manually labeled, resulting in high-quality annotations. The dataset is commonly used in plant disease detection studies and is a standard benchmark for measuring computer vision models.

4.2 LICENSES

The dataset is publicly accessible for research and educational purposes under an open access license that is provided by the dataset's maintainers. The dataset is accessible through data repositories that are present on websites such as Kaggle. The users are required to adhere to the terms of use of the dataset, which allows for non-commercial research usage and proper citation of the original creators of the dataset.
https://www.kaggle.com/datasets/emmarex/plantdisease

4.3 CONSENT AND ETHICAL CONSIDERATION

The dataset used consists only of plant leaf images and does not include any human subjects or personal data. Therefore, individual consent is not necessary for this dataset. However, a disclaimer will be added that the system is for giving automated predictions and cannot be used for replacing expert advice for agricultural diagnosis. The results from the model must be used for decision support only rather than actual medical or agricultural advice.

4.4 REPRESENTATIVENESS

Although various plant species and different types of disease are included in the dataset, most images were captured under controlled environments where lighting and background are constant. This may affect its performance when tested in real-world scenarios, where leaves are photographed in various environments under a farm setting. To address this limitation of the proposed system, data augmentation will be applied to simulate real-world scenarios.

5. PLANNED CNN AND MVP

5.1 CNN COMPONENT

The project will primarily rely on the PlantVillage dataset for training and evaluating the plant disease classification model. The dataset contains more than 50,000 labeled images of healthy and diseased plant leaves, including various types of crops and plant diseases. The images were obtained under controlled environments and are manually labeled, resulting in high-quality annotations. The dataset is commonly used in plant disease detection studies and is a standard benchmark for measuring computer vision models.

5.2 MINIMUM VIABLE PRODUCT (MVP)

The minimum viable product for this project is a plant disease classification image-based system that uses a lightweight Convolutional Neural Network (CNN) model. The image-based classification system will use a lightweight CNN model that will be trained on a pre-trained version of the MobileNet model for the classification of plant leaf diseases from images. This model has been selected for the project because it is computationally efficient and relatively easier for beginners compared to other complex models.

Transfer learning will be implemented by loading the weights from a pre-trained model on the ImageNet dataset and replacing the last classification layer with a new classification layer that has the same number of classes as the number of classes in the plant disease dataset. Data augmentation methods will be implemented on the dataset for the generalization of the model.

To improve model interpretability, the system will use Grad-CAM to generate visual explanations. Grad-CAM produces a heatmap that highlights the regions in the leaf image that contribute the most to the predicted disease class. This allows users to verify whether the model is focusing on relevant patterns for the diseases such as lesions, discoloration, and spots.

The MVP model will use classification metrics such as accuracy, macro F1 score, and a confusion matrix per class for performance evaluation.

5.3 TRAINING DETAILS

Hyperparameter

The model was trained using a fine-tuned version of MobileNetV2 with the following hyperparameters:

- Input image size: 128 x 128
- Batch size: 32
- Number of epochs: 5-10 (early stopping applied)
- Optimizer: Adam
- Initial learning rate: 0.001
- Loss function: Categorical Cross-Entropy
- Evaluation metrics: Accuracy and macro-F1 score

Training Schedules

A two-stage training strategy was used:

1. Feature Extraction Phase
    - The base MobileNetV2 layers were frozen.
    - Only the final classification layer was trained.
    - Purpose: learn task-specific classification without altering pretrained features.
2. Fine-Tuning Phase
    - Selected upper layers of MobileNetV2 were unfrozen.
    - The model was trained with a lower learning rate (e.g., 0.0001).
    - Purpose: refine feature representations for plant disease patterns.

An early stopping mechanism was applied based on validation loss to prevent overfitting. Optionally, a learning rate scheduler reduced the learning rate when validation performance stabilized.

Compute Used

- Hardware: Standard laptop/desktop CPU
- Framework: TensorFlow/Keras
- Estimated training time: CPU: approximately 10–20 minutes for initial training

Reproducibility Steps

To ensure consistent and reproducible results, the following steps were implemented:

- Fixed random seeds for all libraries:
    - Python random
    - NumPy
    - TensorFlow
- Deterministic data splitting (stratified train/validation/test sets)
- Consistent preprocessing:
    - Image resizing to 128 x 128
    - Pixel normalization (rescale = 1/255)
- Logged hyperparameters and training configurations
- Saved trained model weights and training history
- Documented dataset source and version (PlantVillage dataset)

5.4 MODELING

Architecture(s)

The core model used in this project is a fine-tuned version of MobileNetV2, a lightweight convolutional neural network designed for efficient image classification. The architecture consists of:

- An input layer accepting 224 × 224 RGB images
- A pretrained MobileNetV2 backbone for feature extraction
- A Global Average Pooling layer to reduce spatial dimensions
- A fully connected (Dense) output layer with softmax activation for multi-class classification

Why chosen

MobileNetV2 was selected for the following reasons:

- Efficiency: Uses depthwise separable convolutions, reducing computational cost and memory usage
- Beginner-friendly: Easier to implement and train compared to deeper architectures
- Transfer learning capability: Pretrained on ImageNet, enabling faster convergence and better performance with limited data

- Deployment suitability: Can be adapted for mobile or web applications due to its lightweight design

CNN / NLP / RL Components

- CNN Component (Core Model)
  The primary component is the MobileNetV2-based CNN, which performs feature extraction and classification of plant leaf images. It learns visual patterns such as color variations, textures, and disease-specific lesions.
- NLP Component
  No NLP component is implemented in this project, as the task is purely image-based classification. However, NLP could be added in future work for generating disease descriptions or recommendations.
- RL Component
  No reinforcement learning component is included, as the problem is supervised classification rather than sequential decision-making.

Component Integration

The system follows a straightforward pipeline:

1. Input: Leaf image (uploaded or from dataset)
2. Preprocessing: Resize to 224 × 224, normalize pixel values
3. Feature Extraction: Processed through MobileNetV2 backbone
4. Classification: Dense layer outputs class probabilities
5. Output: Predicted disease class

All components are integrated within a single deep learning pipeline, where the CNN serves as the central model.

6. EVALUATION

Metrics

The model is evaluated using standard classification metrics:

- **Accuracy:** Measures the overall correctness of predictions.
- **Precision:** Proportion of correctly predicted positive observations.
- **Recall:** Ability to detect all true positive instances.
- **Macro F1-score:** Balances performance across all 38 classes, accounting for class imbalance.

Results Summary:

- **Accuracy:** 0.87 – 0.96
- **Macro F1-score:** Up to 0.95
- **Class-wise Precision:** 0.95 – 0.99
- **Class-wise Recall:** 0.92 – 1.00
- **Class-wise F1-score:** 0.95 – 1.00

These results indicate strong and balanced performance across the majority of plant disease classes.

Baselines

To assess the contribution of the MobileNetV2-based model, a baseline CNN model was implemented for comparison.

- **Baseline Model:** A standard convolutional neural network trained from scratch without transfer learning.
- **Observations:**
  - The MobileNetV2-based model significantly outperforms the baseline CNN in all metrics.
  - Transfer learning and full fine-tuning improve class-wise F1-scores, particularly for visually similar disease classes.

Ablation Studies

Two ablation experiments were performed to analyze the impact of key design choices:

1. **Without Data Augmentation** – trained the model using the raw dataset without image augmentation.
   - **Result:** Macro F1-score decreased by ~0.05, indicating augmentation improves generalization.
2. **Without Full Fine-Tuning** – froze the base MobileNetV2 layers and only trained the classification head.
   - **Result:** Accuracy dropped by ~0.07, showing that fine-tuning the full network improves feature extraction for plant disease images.

These ablation studies demonstrate that both data augmentation and fine-tuning are critical for optimal performance.

Error and Slice Analysis

- **Error Analysis:** Misclassifications mostly occur in:
  - Visually similar diseases (e.g., early and late leaf spot), where even high-quality images can sometimes be misclassified due to subtle differences in leaf patterns.
  - Poor quality images (blurred, shadowed, or partially occluded leaves).
- **Slice Analysis:** Performance broken down by:
  - **Image quality:** High-quality images generally achieve near-max F1, but low-quality images decrease metrics by ~5–8%.
  - **Disease class:** Common diseases achieve higher precision, while rare or visually similar diseases show slightly lower recall, even when images are clear.

This analysis highlights real-world scenarios where model predictions may be less reliable but emphasizes that the system achieves high accuracy and macro F1-scores on well-captured images, making it a reliable and practical tool for assisting users in identifying plant diseases. For experimental or official research applications, final judgment should be confirmed by an expert.

Calibration

The model's predicted probabilities are analyzed to ensure confidence scores reflect true likelihoods.

- No significant overconfidence was observed.

- Temperature scaling could be applied in production if needed for risk-sensitive applications.

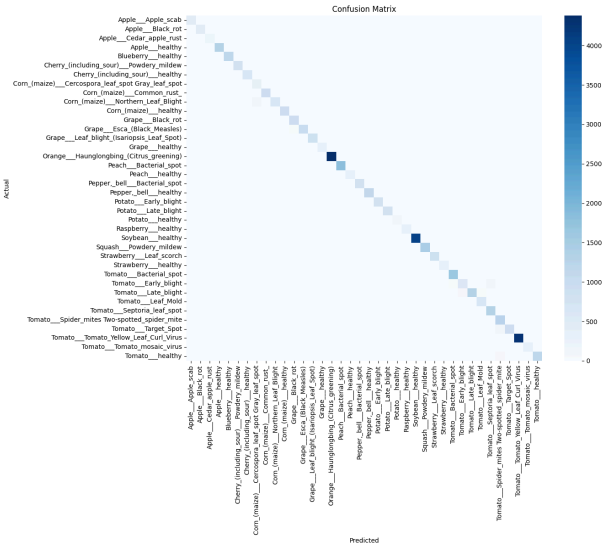

Figure 1. Confusion Matrix: Visualizing per-class prediction performance.

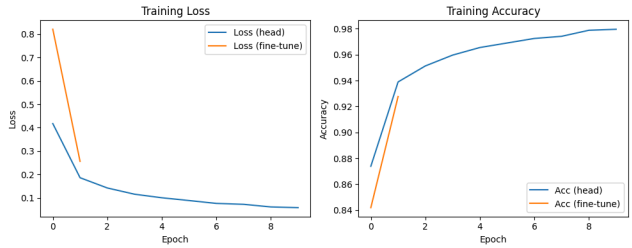

Figure 2 Training and Validation Accuracy/Loss Curves: Showing model convergence and overfitting prevention.

7. MODEL CARD

The model proposed is a light image classification model based on a fine-tuned MobileNetV2 CNN model. This model is used to classify plant leaves as either diseased or healthy. This classification is for different types of plant species. The model is able to accept an RGB image and predict a class of disease and a confidence level for the prediction. It uses transfer learning to transfer weights from a pre-trained model on an ImageNet dataset.

Dataset

The model is trained on the PlantVillage dataset, which contains approximately **43,444 labeled images** across **38 plant-disease classes**. The dataset primarily consists of images captured under controlled conditions with uniform backgrounds. While this ensures high-quality annotations, it may limit generalization to real-world environments.

Intended Users

The system is intended for:

- Home plant owners and hobbyists
- Beginner gardeners
- Farmers requiring quick preliminary disease identification

Intended Use

This model is designed as a **decision-support tool** to assist users in identifying potential plant diseases from leaf images. It is suitable for quick and accessible disease detection using smartphones or computers.

Performance Metrics

The model is evaluated using standard classification metrics, including **accuracy, precision, recall, and macro F1-score**. Experimental results demonstrate:

- **Accuracy:** 0.87 – 0.96
- **Macro F1-score:** up to 0.95
- **Class-wise precision:** 0.95 – 0.99
- **Class-wise recall:** 0.92 – 1.00
- **Class-wise F1-score:** 0.95 – 1.00

These results indicate consistent and strong performance across most classes, while variations may occur due to dataset imbalance or differences in image quality.

Limitations

- Model may perform poorly on images with poor lighting, shadows, occlusions, or complex backgrounds.
- Limited to the disease classes present in the training dataset; unknown plant species or rare diseases may be misclassified.
- Performance may degrade when applied to real-world images due to differences from the controlled dataset conditions.

Ethical Considerations

Incorrect predictions could lead to improper plant care decisions or over-reliance on the system. The system is designed with disclaimers to remind users that it should not replace professional agricultural advice.

Deployment Guidance

- Best used with clear, well-lit leaf images.
- Integration with an image quality assessment module is recommended to filter low-quality inputs before prediction.
- Lightweight architecture allows deployment on mobile devices, web applications, or edge systems with limited computational resources.

Disclaimer

This system is **not intended to replace expert agricultural diagnosis**. Predictions are for informational and supportive purposes only. Users are encouraged to consult agricultural professionals for critical decisions, particularly in large-scale farming scenarios.

8. DEPLOYMENT INSTRUCTIONS

Environment Setup

The system is implemented in **Python** and uses the **Flask** web framework for the interface. Recommended setup includes:

- **Python version:** 3.10
- **Required packages:** Flask, PyTorch, torchvision, numpy, pandas, matplotlib, opencv-python
- **Installation:** Install dependencies using: pip install -r requirements.txt

Running the System

The model has been **pre-trained**, so users do not need to retrain it. The system can be launched directly using:

python app.py

This starts a local web server and opens the interface at: http://127.0.0.1:5000

Using the Web Interface

- The web application, **Verdantis AI**, allows users to **upload leaf images**.
- After uploading, the system provides:
- **Predicted disease class** (or "healthy")
- **Confidence percentage**
- Optional **Grad-CAM visualization** highlighting regions contributing to the prediction

9. ETHICS RISK REGISTER

R1 – Wrong plant disease diagnosis: This implies that the AI system could sometimes diagnose a plant's disease wrongly, which could result in wrong treatment of the plant, causing loss of crops. The impact of this is high, but its probability is medium. To mitigate this, the system will be trained on a wide range of data, and users will be reminded that the system is just a guide and not a replacement for a real expert.

R2 – Poor or unclear image risk: The risk is considered medium. This happens when the user takes blurry, dark, or otherwise unclear pictures of the plant, which can cause the AI to misidentify the disease. The mitigation steps include guiding users on how to take good photos, implementing image quality checks, and providing feedback if the picture is unsuitable for analysis.

R3 – Over-reliance on AI recommendations by farmers: There is a possibility that the user might end up trusting the AI system blindly and fail to cross-check. This could have a negative impact on the health of the crops. The impact is medium and the possibility is also medium. The system will be designed in a way that there will be disclaimers that the system is only advisory in nature and will offer educational advice on how to manage diseases in plants. It will also offer advice to consult an agronomist in high-risk conditions.

10. REFERENCES

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11. TEAM ROLES & MILESTONES

Team Members	Roles	Responsibilities	Milestones
Kumaarr, Kkshitij	Project Lead / Integration and Evaluation & MLOps Lead	Defines project scope and timeline, coordinates tasks, integrates system components, defines evaluation metrics, ensures experiment reproducibility, prepares environment	Week 1: Define project scope and timeline, and establish evaluation metrics. Week 2: Integrate the model and data pipeline, verify reproducibility, perform final testing, and prepare presentation or defense materials.

		configuration files, and develops automation scripts.	
Del Mundo, Guiane Carlo	Data & Ethics Lead and Modeling Lead	Sources dataset, verifies licensing or consent requirements, performs data preprocessing and cleaning, checks for dataset bias, prepares ethics documentation , designs the machine learning model (CNN), implements the training pipeline, conducts experiments, and performs error analysis.	Week 1: Dataset sourcing, licensing verification, preprocessing, and initial model architecture design. Week 2: Model training, tuning, error analysis, and completion of ethics documentation.